Physical Chemistry Cume

Molecular Structure and Bonding

Apr 6th 2012

1) The allyl cation, $CH_2 = CH - CH_2^+$, has a delocalized π network that can be described using the Hückel method. The Hamiltonian matrix within the Hückel approximations is

$$H = \begin{pmatrix} \alpha & \beta & 0 \\ \beta & \alpha & \beta \\ 0 & \beta & \alpha \end{pmatrix}$$

a) Explain the significance of the α and β terms.

b) Are they positively or negatively valued?

25 points

2) a) Derive the MO energy levels for the allyl cation and place the electrons in the energy levels appropriate for the ground state. (**Hint**: start with the secular determinant)
b) Classify the MOs as bonding, antibonding or nonbonding.

45 points

3) Now consider the allyl radical, $CH_2 = CH - CH_2^{\bullet}$, and allyl anion, $CH_2 = CH - CH_2^-$.

a) Using your MO results from the allyl cation, place the electrons in the energy levels appropriate for the ground state.

b) Which of the 3 species (allyl cation, allyl radical and allyl anion) are paramagnetic?

c) What is the bond order for the 3 species?

30 points